

INDIVIDUAL PROJECT (LSDS)

WEEK 5 REPORT **(Apache Pig Implementation)** **Name - Kuunal Mistry** **AppID - 2410130**

1) Objective of Week 5:

To analyze the automobile dataset stored in HDFS using Apache Pig and perform aggregation to compute average price per brand.

2) Dataset Details:

- File: car_price_big.csv
- Size: ~152MB
- Records processed: 2,500,001
- Location: /user/maria_dev/car_project/

3) Pig Script:

Load Data:

```
grunt> car_data = LOAD '/user/maria_dev/car_project/car_price_big.csv'
>> USING PigStorage(',')
>> AS (
>>     car_id:int,
>>     brand:chararray,
>>     year:int,
>>     engine_size:double,
>>     fuel_type:chararray,
>>     transmission:chararray,
>>     mileage:int,
>>     condition:chararray,
>>     price:double,
>>     model:chararray
>> );
```

Remove Header:

```
grunt> clean_data = FILTER car_data BY brand != 'Brand';
```

Group By Brand And Compute Average Price:

```
grunt> grouped = GROUP clean_data BY brand;
grunt> avg_price = FOREACH grouped GENERATE
grunt>     group AS brand,
>>     AVG(clean_data.price) AS avg price;
```

Store Output:

```
grunt> STORE avg_price INTO '/user/maria_dev/car_project/pig_avg_output_final';
2026-03-02 11:16:32,010 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features
used in the script: GROUP_BY,FILTER
2026-03-02 11:16:32,057 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schema
tuple] was not set... will not generate code.
2026-03-02 11:16:32,107 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOpti
mizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParall
elSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, PartitionFilterOpt
imizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, Stream
TypeCastInserter]}
2026-03-02 11:16:32,182 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Select
ed heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usag
eThreshold = 489580128
2026-03-02 11:16:32,309 [main] INFO  org.apache.pig.backend.hadoop.executionengine.tez.TezLau
ncher - Tez staging directory is /tmp/maria_dev/staging and resources directory is /tmp/temp-
521796584
```

4) Execution Summary:

- Successfully read: 2,500,001 records
- HDFS bytes read: ~160MB
- Output records: 7
- Output file size: 165 bytes
- Execution engine: Tez

5) Result:

```
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /user/maria_dev/car_project/pig_avg_output_final
Found 2 items
-rw-r--r--  1 maria_dev hdfs          0 2026-03-02 11:17 /user/maria_dev/car_project/pig_avg_output_final/_SUCCESS
-rw-r--r--  1 maria_dev hdfs        165 2026-03-02 11:17 /user/maria_dev/car_project/pig_avg_output_final/part-v001-o000-r-00000
[maria_dev@sandbox-hdp ~]$ hdfs dfs -cat /user/maria_dev/car_project/pig_avg_output_final/part-v001-o000-r-00000
BMW      54157.11438548624
Audi     51953.424809786135
Ford     51593.25481268938
Honda    52050.283948846
Tesla    53475.5474712718
Toyota   52078.728235304
Mercedes 53191.090084999494
[maria_dev@sandbox-hdp ~]$
```

6) Comparison With Hive & MapReduce:

The Pig results match exactly with:

- Hive query results (Week 4)
- MapReduce output (Week 3)

This confirms consistency across different Hadoop ecosystem tools.

7) Conclusion:

Apache Pig successfully performed large-scale data processing and aggregation using a high-level scripting approach. It simplifies big data analytics compared to writing raw MapReduce programs while still leveraging distributed processing through YARN.

8)Hue Interface Usage:

The current HDP 2.6 sandbox environment does not include the Hue service. Therefore, data exploration and analysis were performed using:

- Apache Pig (command-line interface)
- Apache Hive (CLI interface)
- HDFS file system commands

All required data exploration and query operations were successfully demonstrated through these tools.